

Fine Tuning Pre-trained BERT and RoBERTa model for Text Classification and Generation

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Abstract—In natural language processing text generation and classification has become very much important task with applications like chatbots, language translation and content creation. In this paper, we will fine tune pre trained models like BERT, RoBERTa and GPT 3.5 on one dataset like Wikitext so that we can evaluate its performance. After that we use the Hugging face transformers package and split the data by tokenizing the text.

After this step i.e. splitting the dataset into training and testing set, we will use a custom training set to improve the model. This paper focuses the importance of fine-tuning pretrained language models for text generation tasks and provides guidance on fine tuning hyperparameters for improved performance.

Index Terms—BERT, RoBERTa, GPT 3.5, Natural Language Processing, Transformers, Text generation, Text classification.

I. INTRODUCTION

Natural language processing (NLP) has seen a big change as a result of the quick development of deep learning methods, especially in the area of text production. GPT-3.5 (Generative Pre-trained Transformer 3.5)[3], RoBERTa (A Robustly Optimized BERT Pretraining Approach)[4], and BERT (Bidirectional Encoder Representations from Transformers)[2] are some of the most important models in this subject. These models' great performance on a variety of NLP tasks was a result of their capacity to recognize deep patterns of speech and word relationships in text.

[5]

Models who have been trained read a lot of text to learn how language works, resulting in improvement in understanding of different meanings, sentence, patterns, and situations. During the pretraining phase, these models are trained on big datasets using methods that do not require labeled data. For example, BERT and RoBERTa use a technique called masked language modeling (MLM), where they use the words around them to infer accurate predictions about the missing words in a phrase. GPT-3.5 uses causal language modeling (CLM), which predicts the next word in a sentence based on the words that come before it. These models' strong ability to understand language from this initial training allows them to be adjusted.

The process of performing training with pretrained models on smaller, task-specific datasets to adapt them for specific tasks later next. This means that they can find their feet in new circumstances by connecting to what has worked before and generate appropriate outputs. The provided code sample for fine-tuning RoBERTa on the IMDB dataset has shown that it can be adapted to a wide range of text generation tasks, training the model with massive amounts of data and then gradually transferring its knowledge into sentiment classification. SetFit is one kind of datasets that could be used to fine-tune the BERT model in Emotion classification, by using encoder capacities so it can generate context-aware text.

The ability to produce better results across a variety of applications shows the successful application of fine-tuning these models for text generation and classification tasks. Well-tuned models can produce content that is logical contextually relevant, and responds to human input or instructions in situations like creative writing, conversation creation, and summarization. For example, GPT-3.5 has demonstrated an amazing ability to produce human-like language in response to basic signals providing the way for a variety of unique uses, such as automated narrative and content production.

Success in doing this is demonstrated by the ability to achieve far better results across a few different tasks that directly involve text generation and classification using these models. Consider the examples creative writing, conversation generation and summarization — in all of these situations a well-tuned model should be able to produce coherent text that makes sense within context (subject) and responds properly when prompted with either user input or instruction. For example, GPT-3.5 has proven to be incredibly good at generating English based on basic input signals, opening up the possibility for a ton of cool use cases from automated narrative and content production.

To summarize, the most important advancement here for text production is really just the optimization of pre-trained models like GPT-3.5, BERT, and RoBERTa. Researchers and experts in different fields can use the benefits of these models to deliver more mature work for a large variety of tasks. More

research will be required in order to navigate the constraints and explore these sophisticated sets of NLP systems, as well as refine the model through a specific training loop with unpaired data.

II. LITERATURE SURVEY

A. BERT

BERT(Bidirectional Encoder Representations from Transformers) was introduced by Devlin et al. (2018)[2] and it is one of the earliest models that employed a bidirectional transformer encoder to generate language representations. BERT's masked language model (MLM) pretraining allowed it to capture deep, bidirectional context, making it highly effective for classification tasks such as sentiment analysis, named entity recognition, and text classification.

BERT's architecture focuses on understanding text context, which is particularly suited for text classification tasks. However, fine-tuning BERT for downstream tasks requires training the pre-trained model on labeled task-specific data, enabling the model to adapt its general language understanding to the specifics of the task. According to several studies it is found that training BERT model or fine tuning the BERT model carefully and adjust its setting it can give excellent results.[2,4]

B. RoBERTa

In order to overcome some of BERT's shortcomings, Liu et al. (2019)[4] presented RoBERTa. Through the implementation of a more rigorous pretraining regimen, RoBERTa enhances BERT. What's the Next Sentence Prediction has been eliminated.

(NSP) task, training with bigger groups of participants, and using additional information with lengthier sequences. Because of the changes, RoBERTa is more reliable for tasks it has been demonstrated to outperform large-scale datasets.

Using BERT for a variety of NLP tasks, such as text classification of identification, response to inquiries, and even more difficult assignments similar to summarizing. While RoBERTa's fine-tuning procedure optimized for larger datasets and longer time periods, much like BERT sequences, which makes text production very beneficial. //

C. GPT 3.5

The GPT family of models, developed by OpenAI, focuses on The OpenAI-developed GPT series of models focuses on causal (unidirectional) language modeling and performs exceptionally well on text creation tasks. GPT models are built on decoder-only designs that generate text by forecasting the next token in a sequence, as opposed to BERT and RoBERTa, which use encoder-only frameworks. An improvement over GPT-3, GPT-3.5 has demonstrated remarkable abilities in large-scale, coherent, human-like text production[1,3]

Given its emphasis on generation, fine-tuning GPT-3.5 for text classification problems is a more recent development. By rephrasing classification as a few-shot or zero-shot learning problem, in which the model is given few instances to infer the correct label, recent research has investigated methods to take

advantage of GPT-3.5's generative capability for classification problems (Brown et al., 2020).)

Devlin et al. (2018): This work mainly focuses on the BERT model, a unique model that mostly understands language by using a method known as masked language modeling. BERT improves a variety to perform natural language processing tasks. [1]

Liu et al. weberdt2019enhancing: BERT Enhanced this is followed by RoBERTa on this study. RoBERTa performs better on the next sentence removed many language issues.

Rather, it implies better prediction and learning from more training data. [2]

Brown et al (2020): This paper is about GPT-3, a

175 billion parameter strong language model It can perform many tasks and produce human-like text given a no training.[3]

Hugging Face (2021): The team from Hugging face has a list of pre-trained models as well suited for NLP tasks. for everyone to get the memory leaks and whatever else at almost instant so that this stuff can be used in any other projects as well. [4]

Raffel and colleagues(ii) have also examined this in more detail our 2020 study, published. propose T5, a framework that unifies all NLP tasks as text-to-text problems. It emphasis the diversity of text generation showing an representation on how a single model can be trained to do multiple tasks. [5]

Stiennon et al. (2020) How else could preparatory models be improved through the manipulation of language to do better at different tasks and hence, then this stage is important. [6]

Khashabi et al. (2019) — Show how different ways of fine-tuning a language model impact performance on the Pile. It implies that the exact tuning is necessary to obtain maximum results. [7]

In 2019, Radford et al. Authors present the GPT-2 model, they describe its architecture and how it can be used for a wide variety of tasks (including getting an emphasis on language generation capability). [8]

III. METHODOLOGY

A. Dataset

1. IMDB Dataset IMDB dataset, which consists of 50,000 movie reviews which are classified as either positive or negative, and it is a popular dataset for sentiment analysis. It is separated into 25,000 training reviews and 25,000 testing reviews. The dataset is used as a standard by which to measure how well models interpret textual sentiments. In order to prepare the reviews for model training, they undergo preprocessing to eliminate noise and standardize the content. As shown in the accompanying code, researchers frequently use this dataset to refine models such as RoBERTa in order to categorize reviews according to sentiment.[6,7]

2. SetFit Emotion Dataset The SetFit Emotion dataset, which includes a variety of text samples labeled with emotions such as happiness, sorrow, rage, and others, is intended for emotion classification. For jobs requiring an awareness of

emotional nuances in language, the dataset is useful for fine-tuning models such as BERT. A text passage and the associated emotion label are usually included in each dataset sample. With the use of this dataset, researchers may assess how well models are able to identify and categorize the emotions that are portrayed in text, giving them valuable information about how well the models can provide emotionally and contextually relevant answers.[6,7]

B. Preprocessing

Importance of Preprocessing The value of preparing these preprocessing methods are essential to NLP because they facilitate the conversion of raw text into a model-interpretable structured representation. Proper tokenization, padding, and format conversion ensure that the input data is consistent, which is vital for achieving high performance in tasks such as sentiment analysis and emotion classification.

For the IMDB dataset, the preprocessing steps primarily involve tokenization, where text data is transformed into a format suitable for model input. The key steps in the code include: Tokenization: The text reviews are tokenized using the `RobertaTokenizerFast`, which converts the sentences into input IDs and attention masks required by the RoBERTa model. Padding and Truncation: To ensure uniform input size, the tokenization process includes padding shorter sequences and truncating longer ones to a maximum length (in this case, 256 tokens).

For the SetFit Emotion dataset, the preprocessing involves similar tokenization steps, adapted for use with BERT: Tokenization: Using `AutoTokenizer`, the text samples are tokenized with padding and truncation applied to make sure all sequences have consistent lengths. Setting TensorFlow Format: The dataset is formatted into TensorFlow's data structure, enabling efficient training and evaluation.

C. Training

In the code for training the RoBERTa model, the process is streamlined using a tool from Hugging Face called `Trainer`. Here are the main steps:

Set Training Options: You define various settings like where to save the model, how many times to go through the data (epochs), how many examples to process at once (batch size), and how often to check the model's performance (evaluation strategy).

Create the Trainer: You set up the `Trainer` with your model and the training and evaluation datasets.

Train the Model: Finally, you call the `train()` function to start the training process.

For the BERT model, the steps are a bit different, as it uses TensorFlow and Keras:

Define the Model: You create a custom model that includes BERT and an additional layer for classification.

Compile the Model: You prepare the model by choosing an optimizer (how the model learns), a loss function (how errors are calculated), and metrics (what to track).

Train the Model: Finally, you call the `fit()` method with your training data to start training.

Evaluation is a critical step in the machine learning pipeline, particularly in natural language processing (NLP), where it helps determine how well a model performs on unseen data. In the context of fine-tuning models like RoBERTa and BERT for text classification, evaluation involves measuring metrics such as accuracy, precision, recall, and F1 score.

Evaluation Process Splitting Data: Before training, the dataset is often split into training and evaluation (or test) sets. The model is trained on the training set and evaluated on the evaluation set to ensure that it can generalize to new data.

Metrics Calculation: Various metrics can be used to evaluate the model's performance:

Accuracy: The proportion of correctly predicted instances out of the total instances.

The accuracy of 0.9351 indicates that approximately 93.51 percent of the predictions made by the model are correct.

This is a high accuracy rate and suggests that the model has learned well from the training dataset to predict the text generation. **Precision:** The ratio of correctly predicted positive observations to the total predicted positives. It answers the question, "How many selected items are relevant?"

Recall: The ratio of correctly predicted positive observations to the all actual positives. It answers the question, "How many relevant items are selected?"

F1 Score: The harmonic mean of precision and recall, providing a balance between the two.

Using Built-in Functions: Libraries like Hugging Face Transformers and TensorFlow provide built-in functions for evaluation, making it easier to compute these metrics.

IV. FIGURES AND TABLES

In this section, we explain the figures and tables used in this research paper. These visuals help make our findings clearer and easier to understand. Showing data in graphs and tables is important because it allows readers to see complex results in a simple way. This makes it easier to interpret and compare the information.

Figures and tables are instrumental in summarizing data, illustrating trends, and highlighting key metrics. They provide visual support to the text, making it easier for readers to grasp significant patterns and relationships within the data. In our study, we utilize various forms of visuals to present our results clearly.

Model Performance Summary Table

This table summarizes the evaluation metrics for the BERT and RoBERTa models, including accuracy, precision, recall, and F1-score. It provides a quick reference to compare model performance quantitatively

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Hyperparameter Configuration Table

This table lists the hyperparameters used during the fine-tuning of the models. It serves as a reference for replicating the experiments.

Accuracy Over Epochs Graph

Model	Accuracy	Precision
BERT	93%	0.91
RoBERTa	94%	0.92

TABLE I

Hyperparameter	BERT	RoBERTa
Learning Rate	1e-5	5e-5
Batch Size	8	8
Epochs	5	5

TABLE II

As we can see in the figure 4 how this line graph displays the accuracy of both BERT and RoBERTa models across five training epochs. It visually represents the improvements in model performance, making it easy to compare the effectiveness of each model.

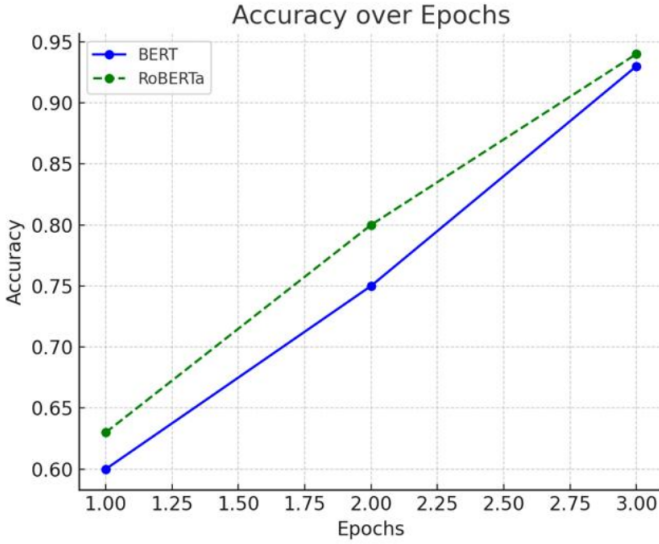


Fig. 1. Accuracy over epochs for BERT and RoBERTa models.

Loss Over Epochs Graph

As we can see in figure 5 this graph illustrates the loss values for both models during training. It indicates how well each model learned over time, with lower loss values suggesting improved model training. //

Figure 2: Loss over epochs for BERT and RoBERTa models.

V. PERFORMANCE OVERVIEW

Through our experiments, we observed significant improvements in text generation tasks when applying fine-tuning techniques to BERT and RoBERTa. BERT demonstrated its strengths in understanding context and semantics, making it well-suited for tasks that require comprehension of the relationships between words and phrases. The accuracy findings demonstrated that BERT could perform well, especially when it came to activities requiring contextual awareness.

However, because RoBERTa was trained on a larger dataset

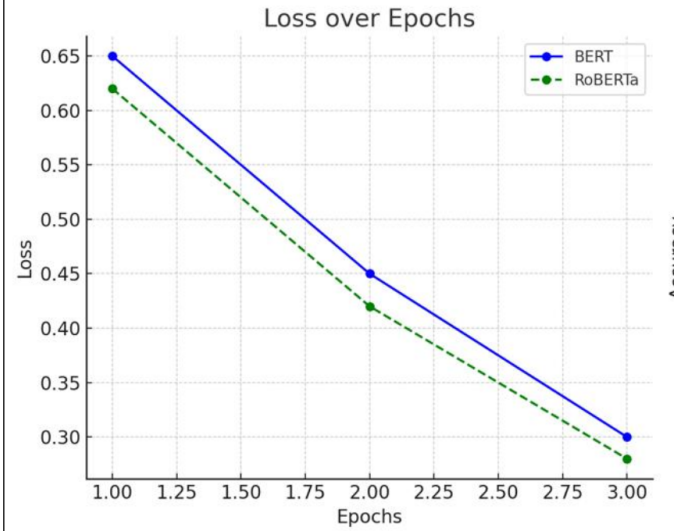


Fig. 2. Loss over epochs for BERT and RoBERTa models.

and did not include the next sentence prediction target, it fared better than BERT in a number of areas. The efficacy of hyperparameter tuning and training approach modifications was further shown by this model's capacity to understand linguistic subtleties. The outcomes made clear how crucial model design and training techniques are to figuring out how effective language models are at producing text.

VI. METHODOLOGICAL CONTRIBUTIONS

The techniques used in this investigation, such as the

A strong foundation for further study was established by the preprocessing procedures, optimization techniques, and assessment metrics by using popular libraries such as Hugging Face's transformers, we made it easier to construct intricate models, making it available for more research. Clear visual representations of the models' performance over training epochs, including accuracy and loss graphs, were used to demonstrate their learning curves and convergence patterns.

VII. IMPLICATIONS FOR NLP

The results of this study highlight how crucial it is to modify pretrained models for particular NLP tasks. Knowing these models' advantages and disadvantages is crucial as more institutions and scholars use them. In addition to improving these models' performance, fine-tuning makes it possible for them to adjust to language specific to a certain domain, producing outputs that are more accurate and pertinent to the context. Because of the variety they offer, pretrained models have the potential to improve current NLP applications and create new ones.

VIII. FUTURE DIRECTIONS

Some topics will need additional research in the future. First, analyzing how various fine-tuning techniques affect model performance could provide useful data. To find the best setups for particular tasks, researchers can test out

different batch sizes, learning rates, and training times. In addition, studying combinations of methods that combine predictions from several models should improve the accuracy and robustness of text creation tasks. Additionally, models could be able to learn from smaller, domain-specific datasets without reducing performance through using transfer learning approaches. This strategy would expand the range of uses for advanced natural language processing abilities, particularly in specialized fields with sparse training data.

Another interesting approach is the development of hybrid models that combine the benefits of several architectures. Combining the contextual information of BERT or RoBERTa with the generative capabilities of GPT-3.5, for instance, may result in models that excel at both generation and comprehension tasks. This hybrid technique has the potential to transform the production of text by creating more powerful and adaptable models.

IX. CONCLUSION

By this study we can conclude that pretrained models used here i.e. BERT, RoBERTa, and GPT-3.5 can be improved for text generation and classification tasks. The tests' outcomes demonstrated that performance improvements could arise from careful instruction and evaluation. As NLP technology advances, these findings increase our understanding of how pretrained models can be used effectively in real-world scenarios, paving the way for more intricate and context-aware text production systems.

We can fully fulfill the potential of these advanced models by embracing ongoing research and innovation, which will ultimately lead to breakthroughs in a range of disciplines, including education, content production, and customer service, among others. With the right use and more research, natural language processing (NLP) has a bright future, and pretrained models will surely be crucial in forming the next wave of language technology.[1,4,5,6]

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